# Vibe-IC — All Steps: Phase → Stage → Step

Every step of the full flow, organised as **Phase → Stage → Step** with a **single continuous numbering 1 → 44** (starting from Stage 1's Spec-to-RTL). Phase 1's document-generation steps are labelled **D1–D5** (pre-flow, not counted in 1→44); two parallel tracks — **Analog A1–A9** and **Mixed-signal M1–M4** — run alongside.

**Phase → Stage map**

- **Phase 1** — Specification & documents: two entries (**Agent path** · **doc-gen path D1–D5**) + optional architecture-exploration front-ends
- **Phase 2** — RTL → synthesis: Stage 1 (RTL+verification) · Stage 2 (constraints+synthesis+DFT+handoff gate)
- **Phase 3** — Physical → Tapeout: Stage 3 (physical+sign-off) · Stage 4 (output+tapeout) · Stage 5 (manufacturing & test)
- **Parallel** — Analog A1–A9 · Mixed-signal M1–M4

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## Phase 1 — Specification & documents

Two entries; both produce the same L-series design documents that feed Phase 2:

- `phase1/input_prompt/` — free text / natural language → **Agent path**
- `phase1/input_doc/` — existing documents / structured YAML → **doc-gen path** (D1–D5)

### Agent path (free-text input)

| Agent | What it does |
|---|---|
| **IC Expert Agent** | The single Phase-1 front door: faces the user directly in plain language, turns natural-language requirements into design facts (one plain-language question per gap), then reviews every layer with silicon expertise, fills sensible defaults, and runs cross-layer consistency checks. |

Flow: user free text → IC Expert Agent → finalised L-series documents.

### doc-gen path D1–D5 (existing-document input)

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| D1 | Doc extraction → L1–L13 | Ingest prompt/docs into `input_doc/` and deterministically extract the core design layers. | user docs / prompt | `L1–L13` JSON | deterministic extractors | `phase1_all_l_docs_present_check` skills: `phase1` |
| D2 | Core design-layer docs L1–L13 | Deterministic extraction of datasheet, FRS, register map, etc. | D1 plain text | `L1_DATASHEET` ··· `L13` | deterministic extractors | — |
| D3 | Extended docs L14–L27 | Protocol, timing, power intent (L21), skeletons. | L1–L13 | `L14–L27` JSON | overlay extractor | — |
| D4 | Protocol-class synthesis | Detect the IC's protocol class (86 classes) and synthesise class facts. | full input text | `ic_class` + protocol facts | is_ + _synth | — |
| D5 | Coverage report | Verify the input documents landed completely in the L docs. | input docs + L docs | parity / coverage report | parity reporter | — |

### Architecture-exploration front-ends (optional, feed Step 1)

| Front-end | What it does | Input | Output |
|---|---|---|---|
| architecture-explore | Design-space exploration (pipeline depth / parallelism / memory vs PPA, Pareto filter). | L docs + perf targets | architecture decisions (feed Step 1) |
| | C/C++/SystemC → RTL via HLS (open-source XLS, etc.). | C/SystemC model | RTL (verified from Step 2 onward) |

| Front-end | What it does | Input | Output |
|---|---|---|---|
| hls-c2rtl | | | |
| SpinalHDL/ Chisel front-end | `eda_spinalhdl_gen` emits Verilog from Scala HDL. | SpinalHDL source | Verilog RTL |

Front-end precedence (**artifact-driven**): existing RTL > C/SystemC model (hls-c2rtl) > SpinalHDL/Chisel > prompt-only (Step 1 spec-to-RTL). The available artifacts decide the path — never pick one speculatively.

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# Phase 2 — RTL → synthesis

## Stage 1 — RTL generation & verification

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| 1 | Spec-to-RTL | Author synthesizable RTL from the L-series docs (SoC/CPU classes may take the IP-catalog reuse + glue path, e.g. Caravel-class harness platforms). | L1–L27 docs | `rtl/*.v(.sv)` • coverage report | — (AI-authored from L docs; SoC via IP-catalog) | skills: `spec-to-rtl` |
| 2 | Lint (RTL hygiene) | Static RTL style/bug checks; auto-fixable issues fixed first. | RTL | lint reports (hygiene / ROM-init) | Verilator lint | `rtl_hygiene_lint` • `rom_init_lint` • `rtl_bug_report_schema_check` • `internal_vs_external_timing_check`…<br>skills: `rtl-review` |
| 3 | CDC / RDC check | Clock-domain / reset-domain crossing safety. | RTL | CDC/RDC reports (crossing / async / reset-dep) | in-house CDC/RDC scan | `cdc_crossing_check` • `cdc_async_input_check` • `reset_dependency_check` |
| 4 | Simulation (testbench + coverage) | Per-IC oracle testbench functional simulation (golden compares) + coverage measurement; when L21 declares power domains, the TB should cover power-state transition scenarios (no open-source UPF-aware sim — structural verification stays with M2). | RTL • L10 test cases | sim logs • results.xml • coverage report | iverilog/vvp • Verilator coverage `eda_simulate` | `testbench_gen` • `coverage_closure` • `l10_tb_conformance_check` • `l12_tb_coverage_check`…<br>skills: `testbench-gen` |
| 5 | Formal verification (assertions) | Prove key properties hold: safety invariants proved unbounded and functional properties bounded-model-checked with the bound disclosed (`all_proved` requires the .sby + SymbiYosys evidence chain). | RTL • L3 constraints | `.sby` • formal results • full-stack TB results | SymbiYosys (ABC pdr / bmc3) `eda_formal` | `formal_property_run` • `assertion_property_check` • `bit_level_full_stack_tb_check` • `formal_proof_evidence_check`<br>skills: `formal-verify` |
| 6 | FPGA early prototype | Pre-synthesis behavioral prototype on FPGA. | RTL • board constraints | `.sof` • map report • FPGA verification audit | Quartus `eda_synth` | `fpga_test_harness_gen` • `debug_first_pass` • `quartus_map_audit` • `fpga_verification_audit` |

## Stage 2 — Constraints, synthesis, DFT & handoff gate

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| 7 | Constraint setup (SDC + PVT matrix) | Author timing constraints (SDC) + PVT corner matrix; power intent modelled via L21 and rendered to UPF (handoff artifact — open-source tools do not | L8 timing • L21 • PDK liberty | `*.sdc` • `pvt_matrix.json` • `<top>.upf` (optional, when L21 declares power domains) | — (UPF emitted by the `l21_to_upf_emit` script, not an EDA engine) | `sdc_syntax_check` • `pvt_matrix_check` • `l21_to_upf_emit` • `upf_syntax_check` |

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| | | consume UPF; structural verification stays with M2). | | | | |
| 8 | SDC validation (incl. derived-clock guard) | Validate constraint correctness, completeness, and exception (false_path/multicycle) justification; manual ICG / register-divided clocks must declare a matching `create_generated_clock` (vacuous PASS without derived clocks — i.e. it passes automatically when the condition does not apply: no divided clocks in the design means the check self-skips). | SDC • L8 • RTL | SDC check report • `derived_clock_sdc.json` | — | `sdc_syntax_check` • `sdc_validator_check` • `sdc_exception_correlation_check` • `derived_clock_sdc_required_check` |
| 9 | Synthesis (Yosys → mapped netlist) | Synthesize RTL and technology-map (dfflibmap + abc -liberty) to the standard-cell netlist. | RTL • SDC • liberty | `synth/netlist.v` • area stats | Yosys + abc `eda_synth` | `synth_wrapper_gen` • `synth_netlist_check` • `provenance_check` skills: `synth-doctor` |
| 10 | Pre-layout STA (multi-corner) | Pre-layout static timing analysis (SS/TT/FF) + post-synth power preview (gate-level vectorless, default toggle rate, disclosed via `analysis_mode`; a different accuracy tier from Step 33's post-layout, optionally VCD-vector, analysis). | netlist • SDC • liberty | pre-PnR timing report + summary • `pre_pnr_power_preview.rpt` | OpenSTA `eda_sta` | `sta_report_check` skills: `sta-review` |
| 11 | DFT insertion (scan chain + ATPG) | Insert scan chains + generate patterns (open-source Fault: scan + stuck-at ATPG + TAP; MBIST/LBIST/compression out of open-source scope). | netlist | scan netlist • ATPG coverage report | Fault (scan+ATPG+TAP) `eda_dft` | `fault_atpg_run` • `dft_atpg_coverage_check` |
| 12 | Post-DFT optimization (resynth / buffering) | Re-optimise timing/area after DFT insertion; the emitted netlist must genuinely retain the scan chain — not merely resolve to a path (2026-08-08: closed a matrix_63x8 dimension-2 gap where an empty file, the pre-DFT netlist copied over, or a netlist whose scan chain silently vanished all satisfied the old files_exist-only gate; Step 13's LEC does not catch scan-chain loss, since scan insertion is designed to be functionally transparent). | scan netlist | `post_dft_netlist.v` | Yosys resynth | `dft_post_optimization_scan_survival_check` skills: `synth-doctor` |
| FS1 | ISO-26262 FMEDA diagnostic-coverage (safety designs only) | Fault-injection on a declared safety mechanism (ECC/parity): inject stuck-at faults on the protected path and measure diagnostic coverage against the ASIL floor. Not applicable for non-safety designs. | RTL • declared safety mechanism • ASIL | fmeda_coverage report (measured DC) | iverilog fault-injection | `fmeda_fault_injection_coverage` • `fmeda_coverage_check` |
| DT1 | Transition-delay-fault (at-speed LOC) ATPG | Generate launch-capture 2-patterns for transition faults on a 2-time-frame launch-on-capture unroll of the scan-cut netlist, and grade the transition test coverage. Not applicable for combinational / no-scan designs. | scan-cut netlist • clock | transition coverage report | Yosys SAT (`sat -prove`) | `transition_fault_atpg_run` • `transition_coverage_check` |
| DT2 | Path-delay-fault (at-speed, timing-graded) ATPG | Report the K longest launch-on-capture paths from the routed netlist with real post-layout timing, then generate and grade a launch-capture 2-pattern per path (robust and non-robust); a path with no possible pattern is excluded, never counted. | routed netlist • SPEF • SDC • scan cut | path-delay coverage report | OpenSTA + Yosys SAT (`sat -prove`) | `path_delay_fault_atpg_run` • `path_delay_coverage_check` |

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| | | Not applicable before the routed netlist and parasitics exist. | | | | |
| DT3 | Small-delay-defect (SDD) at-speed grade | Grade each timing-critical path fault by its real post-layout slack: a defect is caught at-speed only when the detecting path's margin is tight, so detection through a low-slack path is a strong catch and through a slacky path is weak. A slack-rich design honestly scores low (its margin masks small delays). Descriptive grade, no floor. Not applicable before the path-delay and transition coverage exist. | path-delay coverage • transition coverage • SDC | SDD coverage report | OpenSTA + Yosys SAT (`sat -prove`) | `sdd_atpg_run` • `sdd_coverage_check` |
| 13 | Equivalence check (RTL ≡ netlist) | Formally prove gate-level netlist ≡ RTL (LEC). | RTL • post-DFT netlist | LEC report | Yosys equiv | `lec_equivalence_check`<br>skills: `equivalence-check` |
| 14 | Synthesis handoff gate (pre-PnR Yosys audit) | Synthesis→PnR handoff QA: synth script + netlist audit (**open-source-Yosys specific**; the synthesis stage's closing gate). | synth script • netlist | handoff audit reports | Yosys script/netlist audit | `yosys_hilomap_required_check` • `yosys_script_template_check` |

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# Phase 3 — Physical design → Tapeout

## Stage 3 — Physical design & sign-off

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| 15 | Floorplan + PDN | Chip floorplan + power delivery network; tapcell insertion (latch-up well-ties, SKY130 14 µm rule). | netlist • hardmacro LEFs (`pdk_local/` auto-included, via the IP integration check: LEF/GDS/Liberty alignment + corner coverage + L21 supply consistency; macro LEFs should carry obstruction layers) | `floorplan.def` • PDN | OpenROAD (init_floorplan+pdngen+tapcell)<br>`eda_pnr` | `phase3_backend_step` • `floorplan_pdn_check` • `ip_integration_check` |
| 16 | Clock planning | Clock-tree distribution strategy. | floorplan | `clock_plan.json` | OpenROAD CTS planning | `clock_plan_check` |
| 17 | Placement (global + detailed) | Place standard cells. | floorplan • netlist | `placed.def` | OpenROAD (global+detailed place)<br>`eda_pnr` | `placement_legality_check` |
| 18 | Spare-cell + ECO-prep insertion | Pre-place spare cells / ECO readiness so later fixes are metal-only (paired with Step 32 ECO: provisioned here, consumed there). | placed.def | `spare_cells.json` • coverage report | OpenROAD<br>`eda_pnr` | `spare_cell_coverage_check` (preservation is audited at Step 34, after the passes a spare must survive) |
| 19 | CTS (clock tree synthesis) | Build and balance the clock tree. | placed.def • clock plan | `post_cts.def` • clock-tree report | OpenROAD CTS<br>`eda_pnr` | `cts_quality_check` |
| 20 | Post-CTS hold fixing | Repair hold violations after CTS (the runner repeats hold repair post-global-route). | post_cts.def | `post_hold.def` | OpenROAD repair_timing -hold | `hold_closure_check`<br>skills: `hold-fix` |
| 21 | Routing (global + detailed) | Complete all signal routing. | post_hold.def | `routed.def` • router DRC report | OpenROAD TritonRoute<br>`eda_pnr` | `drc_report_check` • `def_stage_progression_check` • `provenance_check` |
| 22 | Parasitic extraction (SPEF) | Extract post-route parasitics; when the PDK ships no coupling captable, lateral coupling caps are added analytically from the routed geometry with a disclosed generic dielectric. | routed.def • tech LEF | SPEF (coupling-aware) | OpenRCX<br>`eda_extraction` | `spef_extraction_check` • `provenance_check` |
| 23 | Post-route STA (multi-corner sign-off) | Sign-off timing with real parasitics (MMMC = per-corner loop, one report per corner). | netlist • SPEF • SDC • multi-corner liberty | post-route timing report • `per_corner/` | OpenSTA (per-corner loop)<br>`eda_sta` | `sta_report_check`<br>skills: `sta-review` |

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| 24 | IR drop (static + dynamic) | Power-grid voltage-drop analysis vs budget: static, plus VCD-vectored dynamic IR (activity-weighted droop under a real switching VCD). | routed.def (PSM) • VCD | static + dynamic IR reports | OpenROAD PSM ( `read_vcd` ) | `ir_drop_report_check` • `dynamic_ir_drop_check`<br>skills: `ir-drop-triage` |
| 25 | EM check (electromigration) | Current-density / metal-lifetime screen. | routed.def (PSM -enable_em) | EM report + per-segment currents | OpenROAD PSM -enable_em | `em_report_check` |
| 26 | Antenna check | Detect + repair process-antenna violations. | routed.def | antenna report | OpenROAD check_antennas/repair | `antenna_report_check` |
| 27 | Signal integrity (crosstalk) | Crosstalk/noise screen (SPEF coupling-cap; advisory tier explicitly named). | SPEF | SI report (incl. >0.9 coupling watch-list) | in-house SPEF coupling screen (OpenSTA window advisory) | `si_crosstalk_check` |
| 28 | PERC / Reliability sign-off (ESD + latch-up + cross-domain) | Enforced sign-off: ESD pad-ring + discharge topology, latch-up well-tap, cross-voltage-domain protection; maps to the four PERC categories — netlist checks + netlist-driven layout checks (automated), current density + P2P resistance (named manual-review). Manual review is signed off by a senior physical-design / reliability engineer; criteria live in `perc_equivalent.json` (categories[].status=MANUAL_REVIEW + `review_criteria`: PDK Jmax tables, foundry ESD discharge-path P2P limit, Vhold>Vdd, L21 cross-domain contract), results back-filled into checklist[].confirmed. | gate netlist • routed.def • L21 power intent • L3 ESD spec • step-24–27 reports | `perc_equivalent.json` • PERC memo • gate verdict | in-house PERC-equivalent (DEF-driven) | `perc_signoff_check` |
| 29 | Post-layout gate-level simulation (SDF) | Gate-level sim with SDF delays to confirm post-layout function (honest SKIP when no SDF re-sim ran). | gate netlist • SDF • TB | post-sim results | iverilog + SDF<br>`eda_simulate` | `post_layout_sim_check` |
| 30 | Post-layout SPICE verification | Transistor-level ngspice correlation vs the Liberty timing: a representative cell plus the top-N STA critical paths, one worst path per distinct endpoint (extracted subckts stitched stage by stage with real net cap; SPICE vs STA path delay per path, aggregated). | SPICE decks • SPEF • STA paths | cell + top-N path SPICE correlation reports | ngspice + OpenSTA<br>`eda_spice` | `spice_correlation_check`<br>skills: `ams-sim` |
| 31 | Physical verification (DRC + LVS + ERC + density) | Sign-off physical rules; density here is **RULE COMPLIANCE** (KLayout per-layer CMP-window deck; execution verification → Step 34, optimization advisory → Step 35); LVS = Magic extraction + real netgen compare (macro-bearing designs — e.g. Caravel-class harnesses — may blackbox macros: Magic `lef write -hide` masks the macro to its interface shell, a netgen supplementary setup compares it as a same-name blackbox; the waiver basis = device-level match + KLayout cross-check, written by `signoff_waiver_emit` into the project's `waivers.json`, cross-referenced as reviewer to-dos by the Step 36 checklist's open_waivers). | GDS • gate netlist • PDK decks | sign-off DRC • LVS • ERC reports | KLayout DRC • Magic ext2spice + netgen LVS • OpenROAD ERC<br>`eda_drc_klayout` • `eda_lvs` | `erc_density_check`<br>skills: `drc-fix` • `lvs-triage` |
| 32 | Post-route timing repair pass | Multi-corner `repair_design` + `repair_timing -setup`, then a FULL `global_route` + `detailed_route` on the | sign-off reports | ECO log or no-ECO flag | OpenROAD ECO | `eco_loop_audit`<br>skills: `eco-plan` |

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| | | routed DEF. NOT an ECO: it re-routes the whole design rather than preserving the implementation, and there is no released revision for a change order to act on. **It does NOT consume the Step 18 spare cells** — measured: zero spare-cell references in the 180-line generator, and no `dont_touch` / `preserve`. Renamed 2026-07-31. | | | | |

## Stage 4 — Output & Tapeout

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| 33 | Power analysis (post-layout) | Full-chip power sign-off (post-layout vectorless OpenSTA report_power; optional VCD vector mode). | netlist • SDC • liberty (+ optional VCD) | power report (leakage+dynamic, analysis_mode) | OpenSTA report_power (+ optional VCD) | `power_report_check` |
| 34 | Metal fill (density fill insertion) | Std-cell-row filler placement (white-space); density here is **EXECUTION VERIFICATION** (the `metal_fill_density_check` gate owns the density verdict; rule compliance → Step 31, optimization advisory → Step 35). | routed.def | `filled.def` • density report | OpenROAD filler_placement `eda_pnr` | `metal_fill_density_check` • `spare_cell_preservation_check` |
| 35 | DFM screen (manufacturability) | Manufacturability screen: redundant-via ratio (single-cut fraction advisory) + density **OPTIMIZATION ADVISORY** (cross-references the Step 34 gate result only — never a duplicate FAIL); OPC/RET/SRAF/PSM as FOUNDRY_SIDE disclosure items (escalated to DESIGNER_COLLAB_REVIEW at ≤28nm; the node is derived by `dfm_screen_check` from the `input/pdk/liberty` filenames and recorded in the same `dfm_screen.json` — process_nm / advanced_node / foundry_side fields). | routed.def • density report | `dfm_screen.json` (via stats + foundry-side list) | in-house DEF via statistics + density cross-ref | `dfm_screen_check` |
| 36 | Tapeout checklist (final sign-off) | Item-by-item final confirmation (substance checks: DRC counts, evidence chains). | all sign-off reports | `tapeout_checklist.json` | — (inventory aggregation) | `tapeout_signoff_check` skills: `tapeout-checklist` |
| 37 | GDSII output | Stream the foundry-deliverable GDSII (only when Step 31 PV is fully clean). | routed.def • merged GDS | sign-off `*.gds` | Magic/KLayout stream-out `eda_gds` | `gds_size_check` • `provenance_check` |
| 38 | Foundry handoff (mask spec + WAT + scribe + corner vectors) | Foundry physical mask kit: mask spec + WAT plan + scribe PCM + corner ATE vectors (chip-specific; foundry-supplied fields named `PENDING_FOUNDRY_*` — tracked in the Step 36 checklist and back-filled after the foundry replies). | GDS • netlist stats • L10 cases | `mask_spec.json` • `wat_plan.json` • scribe • `corner_test_vectors.json` | — (pack generator) | `foundry_handoff_package_check` skills: `tapeout-checklist` |
| 39 | FPGA final sign-off (on-board test) | Final FPGA recompile + on-board attestation with hardware evidence. | RTL • board | final `.sof` • `on_board_pass.json` | Quartus + on-board measurement | `bringup_plan_gen` • `fpga_on_board_attestation_check` |

## Stage 5 — Manufacturing & test (post-fab; triggers on silicon receipt)

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| 40 | Fabrication (foundry, external) | | foundry handoff kit | mask/wafer intake attestations | external foundry | `manufacturing_fab_intake_check` |

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| | | Foundry mask-set + wafer fab (OPC/ RET are foundry-side mask synthesis). | | | | |
| 41 | Wafer sort / probe test | Wafer probing, good-die selection; yield independently re-derived vs target. | wafer lot • probe card | `wafer_sort_yield.json` • `wafer_map.csv` | ATE + probe card (external) | `wafer_sort_yield_check` |
| 42 | Packaging (assembly) | wirebond / FC-CSP / WLCSP assembly. | good dies | `packaging_log.json` | assembly house (external) | `packaging_intake_check` |
| 43 | Final test (ATE + burn-in) | Post-package final test (functional + parametric + burn-in infant-mortality screen). | packaged units • ATE patterns | `final_test_yield.json` • `burn_in_results.json` | ATE (external) | `final_test_attestation_check` |
| 44 | Reliability qualification (HTOL / FIT) | Long-duration HTOL qual (device-hours / failures / FIT attestation; required for automotive/medical grades; consumer MPW may stay dormant = DEFERRED, never blocks tapeout). | HTOL chamber results | `htol_results.json` verdict | HTOL chamber (external) | `htol_attestation_check` |

Out-of-flow lab steps (unnumbered): PFA/EFA (destructive FIB/SEM/EMMI failure analysis) and silicon characterization (shmoo) — data originates from external equipment; the plugin owns the data-analytic root-cause layer (`wafer_map_pattern_classify`, yield-diagnostic).

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# Parallel tracks

## Analog A1–A9 (parallel to Stages 1–3)

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| A1 | Analog spec extraction | Extract per-block analog specs from the L docs. | L5 ADI spec | `spec.json` | deterministic spec extract | `analog_a1_spec_extract_check`<br>skills: `analog-spec-extract` |
| A2 | Topology selection | Choose the circuit topology per spec. | spec.json | `topology.md` | AI topology select | `analog_a2_topology_select_check`<br>skills: `analog-topology-select` |
| A3 | Netlist generation | Generate the SPICE netlist. | topology • PDK models | <block>.sp | xschem netlist<br>`eda_xschem_netlist` | `analog_netlist_pdk_check`<br>skills: `analog-netlist-gen` |
| A4 | Corner sweep (PVT) | PVT corner sweep + Monte-Carlo yield (`mc_yield_pct` $\geq$ 95% gate). | `.sp` • PDK corner libs | `corner_results.json` (executed/ derived counts) | ngspice (corner + MC)<br>`eda_spice_corner` | `analog_a4_corner_sweep_check`<br>skills: `analog-sizing-loop` |
| A5 | Analog layout | Complete the analog layout. | netlist | `layout.mag` / GDS | Magic layout<br>`eda_analog_layout` | `analog_a5_layout_check`<br>skills: `analog-layout` |
| A6 | Block physical verification (per-block DRC + LVS) | Per-block DRC + LVS before merge (catches block-level errors top-level PV would mask). | block GDS • netlist | DRC clean / LVS match flags | Magic DRC + netgen LVS | `analog_a6_block_pv_check`<br>skills: `drc-fix` • `lvs-triage` |
| A7 | Post-layout resimulation | Re-simulate with extracted parasitics; compare pre-vs-post specs (>10% degradation loops to A3). | layout parasitics • spec | `pre_vs_post.json` | Magic extraction + ngspice<br>`eda_spice_corner` | `analog_pre_vs_post_layout_check`<br>skills: `analog-extraction-resim` |
| A8 | Hardmacro generation (LEF + Liberty + GDS + Verilog) | Package the block for Stage-3 consumption (LEF should carry obstruction layers — Magic `lef write -hide` or abstract with obs — so top-level routing cannot enter the macro). | layout • characterisation | 4-file hardmacro | Magic/abstract + characterisation | `analog_hardmacro_check`<br>skills: `analog-hardmacro-gen` |
| A9 | Co-simulation / HW verification | Mixed digital+analog simulation and hardware-in-the-loop verification. | hardmacro • digital RTL | cosim / HW measurement results | iverilog + ngspice co-sim<br>`eda_spice` • `eda_simulate` | `mixed_signal_cosim_check` • `analog_hw_spice_correlation_check`<br>skills: `mixed-signal-cosim` • `analog-hw-tuning-loop` |

### Mixed-signal M1–M4 (triggered when analog blocks exist)

Trigger timing: runs once **A8 hardmacros are complete** and **Stage 3 is near closure** (routed/GDS mergeable) — hardmacros must exist before floorplan, but M1's GDS merge waits for the digital side to finish routing. If a hardmacro slips until Stage 3 has entered Step 31, M1 waits for the final hardmacro GDS and Step 31's top-level LVS re-runs (incremental — scoped to the macro-interface change).

| # | Step | What it does | Input | Output | Tools (EDA) | Programs / Skills |
|---|---|---|---|---|---|---|
| M1 | Top-level integration (A+D GDS merge) | Merge digital + analog GDS, place macros; run top-level LVS on the merged GDS (Magic extraction + netgen, macros blackboxed). | digital GDS • hardmacro GDS | `top_merged.gds` • merge/LVS report | KLayout merge + Magic/netgen top-LVS | `mixed_signal_top_lvs_run` (producer) • `mixed_signal_merge_check` (gate)<br>skills: `analog-flow-orchestrate` |
| M2 | Power domain verification (level shifter / isolation) | Structural verification of cross-domain level-shifters / isolation, plus a cross-power-domain signal-crossing check that derives the isolation/level-shifter requirement from the power-domain definitions and audits the UPF strategy. | L21 • merged design | power_domain / level_shifter / isolation / signal-crossing reports | structural check programs | `power_domain_crossing_check` • `level_shifter_required_check` • `isolation_cell_required_check` • `power_domain_signal_crossing_check`<br>skills: `ir-drop-triage` |
| M3 | Mixed-signal verification (AMS co-sim) | AMS co-simulation + interface signal integrity. | merged design • cosim TB | cosim results • interface SI report | AMS co-sim | `mixed_signal_cosim_check` • `mixed_signal_interface_si_check`<br>skills: `mixed-signal-cosim` • `ams-sim` |
| M4 | Mixed-signal sign-off | Top-level verdict rolled up from M1–M3. | M1–M3 reports | `signoff.json` | rollup verdict | `mixed_signal_signoff_check`<br>skills: `tapeout-checklist` |

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## Totals

| Phase | Stages | Steps |
|---|---|---|
| Phase 1 — Specification & documents | two entries (Agent • doc-gen) + architecture front-ends | D1–D5 + IC Expert Agent |
| Phase 2 — RTL → synthesis | Stage 1 • Stage 2 | 1–14 |
| Phase 3 — Physical → Tapeout | Stage 3 • Stage 4 • Stage 5 | 15–44 |
| Parallel | Analog • Mixed-signal | A1–A9 • M1–M4 |

**44 sequential steps** (Stage 1: 1–6 • Stage 2: 7–14 • Stage 3: 15–32 • Stage 4: 33–39 • Stage 5: 40–44), plus Phase 1 (Agent path & doc-gen path D1–D5) and the two parallel tracks (Analog A1–A9 • Mixed-signal M1–M4). Preflight: P0 (environment health check). Conditional lettered steps: FS1 (ISO-26262 FMEDA diagnostic-coverage, safety designs only) • DT1 (transition-delay-fault ATPG, scan designs only) • DT2 (path-delay-fault at-speed ATPG, scan designs after routing) • DT3 (small-delay-defect at-speed grade, after DT2). Orchestrator `vibe_ic_one_shot_runner.py` runs Phase 1 → Phase 2 → Analog → Phase 3.

Out of scope (declined with reasons): designer-EXECUTED OPC/RET (mask synthesis is foundry-side — surfaced as FOUNDRY_SIDE items in Step 35 + noted at Step 40), commercial hardware emulators (FPGA path covers the intent), MBIST/LBIST/EDT compression (no open-source engines), BSR/BSDL, automatic clock gating (no characterized ICG cell in sky130; manual RTL clock gating remains available — the SDC declaration of divided/gated clocks is guarded by Step 8's `derived_clock_sdc_required_check`), via-doubling/CAA (commercial DFM).

**E1–E3 external-lab steps (reserved numbering, outside the 44; activated on automotive/medical grade upgrades):**

| # | Step | Notes |
|---|---|---|
| E1 | PFA / EFA | FIB / SEM / EMMI failure analysis (external lab) |
| E2 | Silicon characterization | shmoo plots • voltage/frequency sweeps |
| E3 | Temperature cycling / mechanical stress | JESD22-class environmental stress (external lab) |

## Appendix: measured wall-clock reference (for intuition)

Numbers come from REAL local sky130 open-source-flow runs (the `duration_s` fields of `reports/orchestrator/*_one_shot.json`; design sizes 0.5k–21k cells) — measured, never estimated. Durations vary with design, constraints and machine; nonlinear in cell count.

| Stage | Measured range | Samples |
|---|---|---|
| Step 9 Synthesis (Yosys) | ~0.4 s – 21 s | 0.5k cells (lpc) → 20k cells (cv32e40p) |
| Steps 15–22 PnR (OpenROAD, full) | ~3 s – 31 min | 3.4k cells (subservient) → 21k cells (sha256) |
| Step 37 GDS write | ~2 – 4 s | all samples |
| Step 31 DRC (KLayout sky130 deck) | ~8 – 84 s | 0.5k → 20k cells |
| Step 31 LVS (Magic + netgen) | this batch's samples took the light/skip path, so no representative figure; with real macro compares (e.g. Caravel-class harnesses) it runs **minutes to hours**, scaling with macro count (a field run motivated the 4-hour timeout) | — |
| Steps 40–43 manufacturing (fab/ sort/ pkg/ final test) | external, weeks-scale | no local measurement |

Licensing & IP: the whole flow relies on open-source tools only (Apache-2.0 project; commercial-tool firewall and output ownership: repo-root `README.md` § IP ownership, plus the DCO/patent pledge in `CONTRIBUTING.md`).

繁體中文版：`ALL_STEPS_v1.4.14.zh-TW.md`.